```
C:\Users\hp\PycharmProjects\PythonProject1\.venv\Scripts\p
   worker_id  total   z-score  probability
0      wr042  360.0  0.503408    69.266140
1      wr042  360.0  0.503408    69.266140
2      wr042  360.0  0.503408    69.266140
3      wr003  442.0  0.880737    81.076979
4      wr010  552.0  1.386909    91.726518
...      ...    ...       ...         ...
9126   wr039  324.0  0.337752    63.222498
9127   wr026  255.0  0.020244    50.807570
9128   wr048  304.0  0.245721    59.705081
9129   wr025  546.0  1.359299    91.297410
9130   wr012  150.0 -0.462920    32.171084

[9131 rows x 4 columns]
```

```
# h0=There is no significant difference between the sales of Baki and Quba
# h1=There is a significant difference between the sales of Baki and Quba.
baki=merged[merged['region']=='Baki']

quba=merged[merged['region']=='Quba']
t_stat,p_value = stats.ttest_ind(baki['total'], quba['total'], equal_var=False)
if p_value < 0.05:
    print('h0 is rejected-There is a significant difference between the sales of Baki and Quba.')
else:
    print('h0 is accepted-There is no significant difference between the sales of Baki and Quba')

#
```

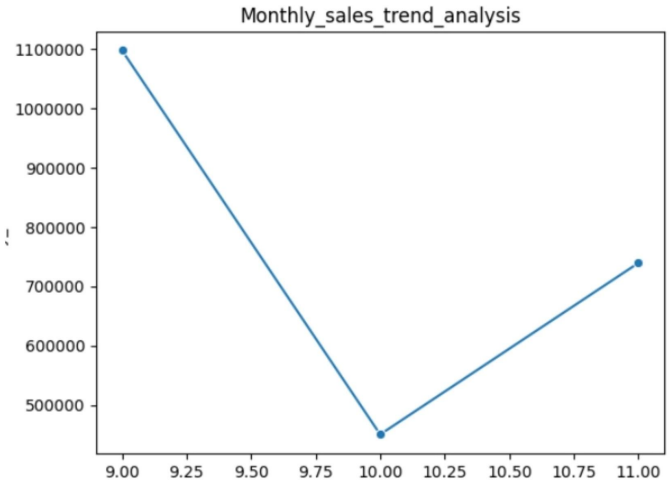
```
import scipy.stats as stats


total_mean=merged['total'].mean()
st_devation=merged['total'].std(axis=0)
merged['z-score']=(merged['total']-total_mean)/st_devation

high_limit=st_devation*3+total_mean
low_limit=st_devation*(-3)+total_mean
satislar2=merged[(merged['total']>high_limit) | (merged['total']<low_limit)]
merged[('probability')]=stats.norm.cdf(merged['z-score'])*100
print(merged[['worker_id','total','z-score','probability']])
```

```
C:\Users\hp\PycharmProjects\PythonProject1\.venv\Scripts\python.exe C:\Users\hp\Pychar
h0 is accepted-There is no significant difference between the sales of Baki and Quba

Process finished with exit code 0
```

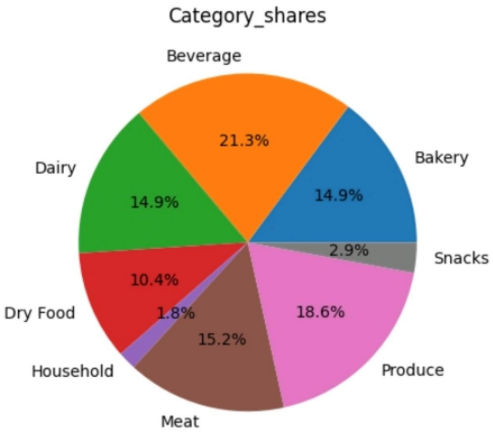
```
C:\Users\hp\PycharmProjects\PythonProject1\
  worker_id    total
41    wr042  380994.0
45    wr048   28932.0

Process finished with exit code 0
```

```
merged['year']=merged['date'].dt.year
aylig_satis=merged.groupby(['year','month']).agg(monthly_sales=("total","sum")).reset_index()
# print(aylig_satis)
ax=sns.lineplot(data=aylig_satis,x=aylig_satis['month'],y=aylig_satis['monthly_sales'],marker='o')
plt.title("Monthly_sales_trend_analysis")
plt.ticklabel_format(style='plain', axis='y')
plt.show()
```

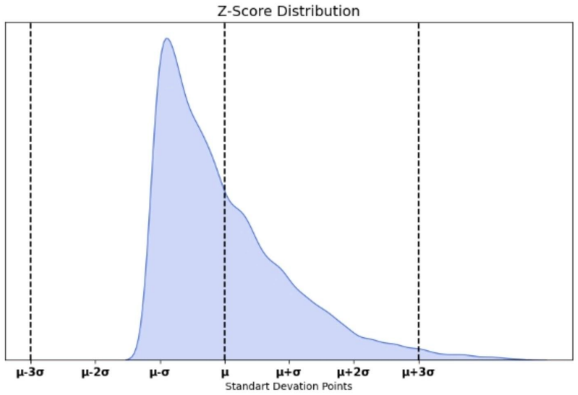
```
sales=merged.groupby('worker_id').agg(total=("total","sum")).reset_index()
quantile1=sales['total'].quantile(0.25)
quantile3=sales['total'].quantile(0.75)
iqr=quantile3-quantile1
outlier1=quantile1-1.5*iqr
outlier2=quantile3+1.5*iqr
outliers=sales[(sales['total']<outlier1) | (sales['total']>outlier2)]
print(outliers)
```



```python
# # vizual3
#
categories=merged.groupby('category').agg(sales=("total","sum"),).reset_index()
categories['pay_faizi']=((categories['sales']/categories['sales'].sum())*100).round(2)
# print(kategoriya_sales)

plt.pie(categories['pay_faizi'],labels=categories['category'],autopct='%1.1f%%')
plt.title('Category_shares')
plt.show()
#
```




```python
sns.kdeplot(merged[('probability')],color='red')
plt.figure(figsize=(10, 6))
sns.kdeplot(merged['z-score'], fill=True, color="royalblue", bw_adjust=0.8)
plt.axvline( x: 0,color='k',linestyle='--',label='mean')
plt.axvline( x: 3,color='k',linestyle='--',label='+3sigma')
plt.axvline(-3,color='k',linestyle='--',label='-3sigma')
plt.yticks([])
plt.ylabel('')
ticks = [-3, -2, -1, 0, 1, 2, 3]
labels = ['μ-3σ', 'μ-2σ', 'μ-σ', 'μ', 'μ+σ', 'μ+2σ', 'μ+3σ']
plt.xticks(ticks, labels, fontsize=11, fontweight='bold')
plt.title( label: 'Z-Score Distribution', fontsize=14)
plt.xlabel('Standart Devation Points')
plt.show()
#
#
```

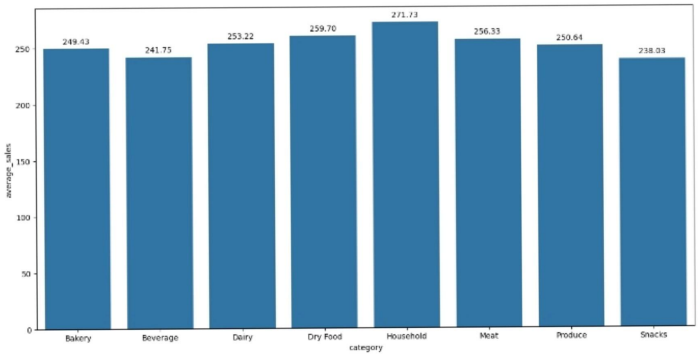
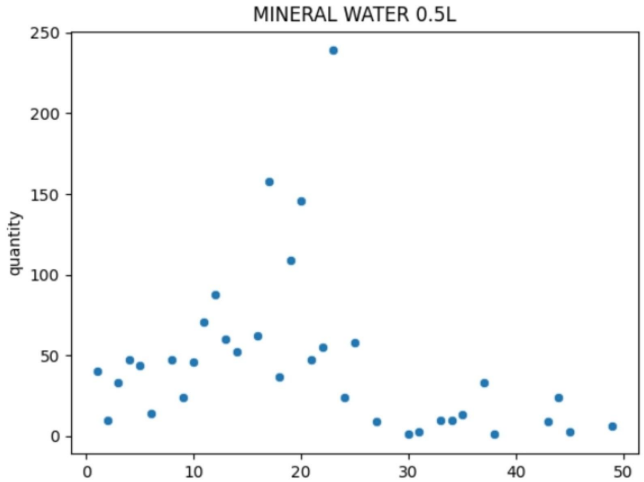


```
#

grouping=merged.groupby('product_id').agg(sales=("total","sum")).reset_index()
grouping['cumulative_percent']=grouping['sales'].cumsum()/grouping['sales'].sum()*100
seksen=grouping[grouping['cumulative_percent']<80]
print(seksen.reset_index())
# # cumulative analysis- first 80% sales
```

```
# # BARPLOT
#
regionlar=merged.groupby('region').agg(total_sales=("total","sum")).reset_index()
plt.ticklabel_format(style='plain', axis='y')
ax=sns.barplot(x='region', y='total_sales', data=regionlar)
for p in ax.patches:
    ax.annotate(format(p.get_height(), '.2f'),
                xy= (p.get_x() + p.get_width() / 2., p.get_height()),
               ha = 'center', va = 'center',
               xytext = (0, 9),
               textcoords = 'offset points')
plt.show()
```

	index	product_id	sales	cumulative_percent
0	0	pr001	22058.0	0.963975
1	1	pr002	32024.0	2.363482
2	2	pr003	17105.0	3.111002
3	3	pr004	25887.0	4.242311
4	4	pr005	18894.0	5.068013
..	...	...	...	...
74	74	pr075	24474.0	75.487691
75	75	pr076	21439.0	76.424614
76	76	pr077	21031.0	77.343707
77	77	pr078	22588.0	78.330844
78	78	pr079	25288.0	79.435976

[79 rows x 4 columns]



```
#
#
beverage=merged[merged['category']=="Beverage"]
beverage_grouping=(beverage.groupby(['product_id','product_name','price'])
            .agg(quantity=('count','sum')).reset_index())
chosen=beverage_grouping[beverage_grouping['product_id']=='pr046']
ax=sns.scatterplot(x='price',y="quantity",data=chosen)
plt.title(chosen['product_name'].iloc[0])

plt.show()
```